

Sprint 5 – Building Business Logic with Apex Placement Management System – Project Submission

Implemented Features

- Retrieved Student and Job records using SOQL
- Prevented duplicate application submissions
- Validated CGPA, Branch, Active Backlogs, and Application Closing Date
- Created Application records using DML
- Updated Application Status
- Developed a comprehensive Apex Test Class with 94% code coverage

Included Screenshots

This report contains the following screenshots:

- ApplicationService Apex Class
- ApplicationServiceTest Apex Class
- Apex Test Execution Results
- Code Coverage Report

```
1 public with sharing class ApplicationService {
2
3     public static String submitApplication(Id studentId, Id jobId) {
4         Student__c student = [
5             SELECT Id, Name, CGPA__c, Branch__c, Active_Backlogs__c, Graduation_Year__c
6             FROM Student__c
7             WHERE Id = :studentId
8             LIMIT 1
9         ];
10        Job__c job = [
11            SELECT Id, Name, Minimum_CGPA__c, Eligible_Branch__c,
12            Backlog_Limit__c, Closing_Date__c
13            FROM Job__c
14            WHERE Id = :jobId
15            LIMIT 1
16        ];
17
18        List<Application__c> existingApplications = [
19            SELECT Id
20            FROM Application__c
21            WHERE Student__c = :studentId
22            AND Job__c = :jobId
23        ];
24
25        if (existingApplications.isEmpty()) {
26            return 'Duplicate Application! Student already applied.';
27        }
28
29        if (student.CGPA__c < job.Minimum_CGPA__c) {
30            return 'Application Rejected: Minimum CGPA not satisfied.';
31        }
32
33        if (student.Branch__c != job.Eligible_Branch__c) {
34            return 'Application Rejected: Branch not eligible.';
35        }
36
37        if (student.Active_Backlogs__c > job.Backlog_Limit__c) {
38            return 'Application Rejected: Too many active backlogs.';
39        }
40
41        if (Date.today() > job.Closing_Date__c) {
42            return 'Application Rejected: Application deadline is over.';
43        }
44
45        Application__c application = new Application__c();
46
47        application.Student__c = studentId;
48        application.Job__c = jobId;
49        application.Application_Date__c = Date.today();
50    }
```

```

33 if (student.Branch__c != job.Eligible_Branch__c) {
34     return 'Application Rejected: Branch not eligible.';
35 }
36
37 if (student.Active_Backlogs__c > job.Backlog_Limit__c) {
38     return 'Application Rejected: Too many active backlogs.';
39 }
40
41 if (Date.today() > job.Closing_Date__c) {
42     return 'Application Rejected: Application deadline is over.';
43 }
44
45 Application__c application = new Application__c();
46
47 application.Student__c = studentId;
48 application.Job__c = jobId;
49 application.Application_Date__c = Date.today();
50 application.Status__c = 'Applied';
51
52 try {
53     insert application;
54
55     return 'Application Submitted Successfully';
56 } catch (Exception e) {
57     return e.getMessage();
58 }
59
60 }
61
62 }
63
64
65 public static String updateApplicationStatus(Id applicationId, String status) {
66
67     Application__c app = [
68         SELECT W.Status__c
69         FROM Application__c
70         WHERE Id = :applicationId
71         LIMIT 1
72     ];
73
74     app.Status__c = status;
75
76     update app;
77
78     return 'Application Status Updated Successfully';
79 }
80
81 }

```



SETUP

Apex Classes

```

1  @isTest
2  public class ApplicationServiceTest {
3
4      @isTest
5      static void testSubmitApplication() {
6
7          // Create Student
8          Student__c student = new Student__c(
9              Name = 'Rahul',
10             Email__c = 'rahul@gmail.com',
11             CGPA__c = 8.5,
12             Branch__c = 'CSE',
13             Active_Backlogs__c = 0,
14             Graduation_Year__c = 2027
15         );
16         insert student;
17
18         // Create Job
19         Job__c job = new Job__c(
20             Name = 'Software Engineer',
21             Company__c = 'TCS',
22             Minimum_CGPA__c = 7.5,
23             Eligible_Branch__c = 'CSE',
24             Backlog_Limit__c = 0,
25             Closing_Date__c = Date.today().addDays(10)
26         );
27         insert job;
28
29         Test.startTest();
30
31         String result =
32             ApplicationService.submitApplication(student.Id, job.Id);
33
34         Test.stopTest();
35
36         System.assertEquals(
37             'Application Submitted Successfully',
38             result
39         );
40     }
41
42     @isTest
43     static void testDuplicateApplication() {
44
45         Student__c student = new Student__c(
46             Name = 'Student1',
47             Email__c = 'student1@gmail.com',
48             CGPA__c = 9,
49             Branch__c = 'CSE',
50             Active_Backlogs__c = 0,
51             Graduation_Year__c = 2027
52         );
53         insert student;
54
55         Job__c job = new Job__c(
56             Name = 'Developer',
57             Company__c = 'Infosys',
58             Minimum_CGPA__c = 7,
59             Eligible_Branch__c = 'CSE',
60             Backlog_Limit__c = 0,
61             Closing_Date__c = Date.today().addDays(5)
62         );
63         insert job;
64
65         ApplicationService.submitApplication(student.Id, job.Id);
66
67         String result =

```


Application Test Execution

To run one or more Apex or flow unit tests, click **Select Tests**. To view the current Apex code coverage for an individual class or your organization, go to the [Apex Classes](#) page.

 In Winter '26 and later, this page shows test details for multiple categories of tests, including Apex and flow tests. For more info about how to [Test Your Changes](#), go to [Salesforce Help](#).

[Select Tests...](#) [Developer Console](#) [Options...](#) [View Test History](#)

Abort			
☐	Status	Class	Result
Test Run: 2026-08-03 11:59:46, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☐		[View] ApplicationServiceTest	
Test Run: 2026-08-03 11:34:33, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (7/7) Test Methods Passed	
Test Run: 2026-08-03 11:34:26, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (7/7) Test Methods Passed	
Test Run: 2026-08-03 11:30:10, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (2/2) Test Methods Passed	
Test Run: 2026-08-03 11:29:53, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (2/2) Test Methods Passed	

Code coverage



Apex Class

ApplicationService

[Back to List](#)

Apex Class Detail

[Edit](#) [Delete](#) [Download](#) [Security](#) [Show Dependencies](#)

Name ApplicationService		Status Active
Namespace Prefix		Code Coverage 94% (32/34)
Created By Jaya Sri Padilam , 03/08/2026, 10:33 am		Last Modified By Jaya Sri Padilam , 03/08/2026, 11:54 am

[Class Body](#) [Class Summary](#) [Version Settings](#) [Trace Flags](#)

Application Record

Student

Swathi

New ContactEditNew Opportunity

RelatedDetails

Student Name

Swathi

Email

swathi@gmail.com

CGPA

8.50

Branch

CSE

Active Backlogs

0

Graduation Year

2,027

Created By

Jaya Sri Padilam, 03/08/2026, 10:31 am

Owner

Jaya Sri Padilam

Last Modified By

Jaya Sri Padilam, 03/08/2026, 10:31 am

Job record

Job

Software Engineer

New ContactEditNew Opportunity

RelatedDetails

Job Title

Software Engineer

Company

TCS

Minimum CGPA

7.50

Closing Date

31/08/2026

Eligible Branch

CSE

Backlog Limit

0

Created By

Jaya Sri Padilam, 03/08/2026, 10:31 am

Owner

Jaya Sri Padilam

Last Modified By

Jaya Sri Padilam, 03/08/2026, 10:31 am

Execute Anonymous output

Developer Console - Google Chrome

resourceful-narwhal-rffxm7-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >

9/07/2026, 10:25:00 AppointmentFuture.apxc Log executeAnonymous @8/3/2026, 11:40:21 AM Log executeAnonymous @8/3/2026, 11:45:05 AM

Execution Log

Timestamp	Event	Details
11:45:05:059	USER_DEBUG	[4] DEBUG Duplicate Application! Student already applied.

☐ This Frame ☐ Executable ☒ Debug Only ☐ Filter

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
Jaya Sri Padilam	Unknown	/services/data/v67.0/t...	8/3/2026, 11:45:05 AM	Success		6.87 KB